Task 4 – Fault State Machine with Structured Recovery

Modular Battery Management System (BMS) – ESP32 / Wokwi

Submission evidence: final Wokwi simulation video, extracted screenshots, structured fault logs, and recovery verification.

1. Objective

The objective of Task 4 is to implement a deterministic fault-management engine with four operating states: NORMAL, DEGRADED, FAILSAFE and SHUTDOWN. The design must identify the source of faults, preserve the active fault identity, record state transitions in a structured format, and use a verification phase before returning from FAILSAFE to NORMAL.

2. System Architecture

The Task 4 implementation is layered on the existing Task 3 Wokwi BMS circuit. The same ESP32, cell potentiometers, SOC/current inputs, relay, LEDs, buzzer, LCD and four pushbuttons are retained. The software adds a state machine and fault supervision layer without requiring a circuit rebuild.

Element	Implementation
Controller	ESP32
State machine	enum: NORMAL, DEGRADED, FAILSAFE, SHUTDOWN
Fault tracking	Fault ID + fault source
Relay monitoring	Relay command vs simulated feedback/mismatch test
ADC monitoring	Frozen ADC test and cell range supervision
Communication	COMM_FAIL / COMM_OK / heartbeat mechanism
Recovery	Verification period before FAILSAFE -> NORMAL
Display	16×2 I2C LCD
Safety outputs	Relay, red/green LEDs, buzzer

3. State Definitions and Transition Table

Current	Event	Next	Action
NORMAL	Communication fault	DEGRADED	Log F003; maintain controlled operation
NORMAL	ADC frozen	FAILSAFE	Log F004; disable relay
NORMAL	ADC/range fault	FAILSAFE	Log F005; disable relay
NORMAL	Relay mismatch	FAILSAFE	Log F002; disable relay
DEGRADED	Critical fault	FAILSAFE	Escalate and enter safe state
DEGRADED	Verified clear	NORMAL	Restore normal operation
FAILSAFE	Fault persists	FAILSAFE	Remain safe; relay off
FAILSAFE	CLEAR + verification passes	NORMAL	Restore normal operation
FAILSAFE	Shutdown request	SHUTDOWN	Terminal safe state
SHUTDOWN	Any event	SHUTDOWN	Remain shut down

4. Fault Identification

Fault ID	Source	Meaning	Safety response
F000	NONE	No active fault	Normal operation
F001	CELL	Battery-cell related fault	Fault tracking / escalation
F002	RELAY	Relay command/feedback mismatch	FAILSAFE
F003	COMM	Communication failure	DEGRADED; escalate if required
F004	ADC_FROZEN	ADC value frozen during injected test	FAILSAFE
F005	ADC_RANGE	ADC/cell reading out of range	FAILSAFE
F006	VOLT_JUMP	Sudden voltage-jump test	FAILSAFE

5. Structured Transition Logging

Every state transition is recorded with a timestamp, previous state, new state, fault ID and fault source. The final demonstration visibly shows records such as:


```
{"timestamp":34654,"previous_state":"NORMAL","new_state":"DEGRADED","fault_id":"F003","fault_source":"COMM"}
```

```
{"timestamp":59404,"previous_state":"DEGRADED","new_state":"FAILSAFE","fault_id":"F002","fault_source":"RELAY"}
```

```
{"timestamp":101078,"previous_state":"FAILSAFE","new_state":"NORMAL","fault_id":"F000","fault_source":"NONE"}
```

```
{"timestamp":117904,"previous_state":"DEGRADED","new_state":"FAILSAFE","fault_id":"F004","fault_source":"ADC_FROZEN"}
```


6. Recovery Verification

The implementation does not immediately return from FAILSAFE to NORMAL when CLEAR is received. A recovery-verification phase starts first. During verification the system remains in the safe condition, checks that injected fault sources are clear, and waits for the verification interval to complete. Only after the checks remain valid is the transition FAILSAFE → NORMAL logged. The final video visibly captures 'RECOVERY VERIFICATION', 'RECOVERY VERIFIED' and 'SYSTEM RETURNED TO NORMAL'.

7. Test Cases and Results

ID	Test	Expected	Observed evidence	Result
TC-01	Startup / NORMAL	NORMAL state	Video + Screenshot 1	PASS
TC-02	COMM_FAIL	NORMAL → DEGRADED, F003	Video + Screenshot 2	PASS
TC-03	RELAY_FAIL	DEGRADED → FAILSAFE, F002	Video + Screenshot 3	PASS
TC-04	CLEAR recovery	Verification then NORMAL	Video + Screenshot 4	PASS
TC-05	FREEZE_ADC	FAILSAFE, F004	Video + Screenshot 5	PASS
TC-06	SHUTDOWN	Terminal SHUTDOWN	Not in supplied video	Not evidenced
TC-07	RANGE	FAILSAFE, F005	Not in supplied final video	Not evidenced
TC-08	JUMP	FAILSAFE, F006	Not in supplied final video	Not evidenced

8. Screenshot Evidence

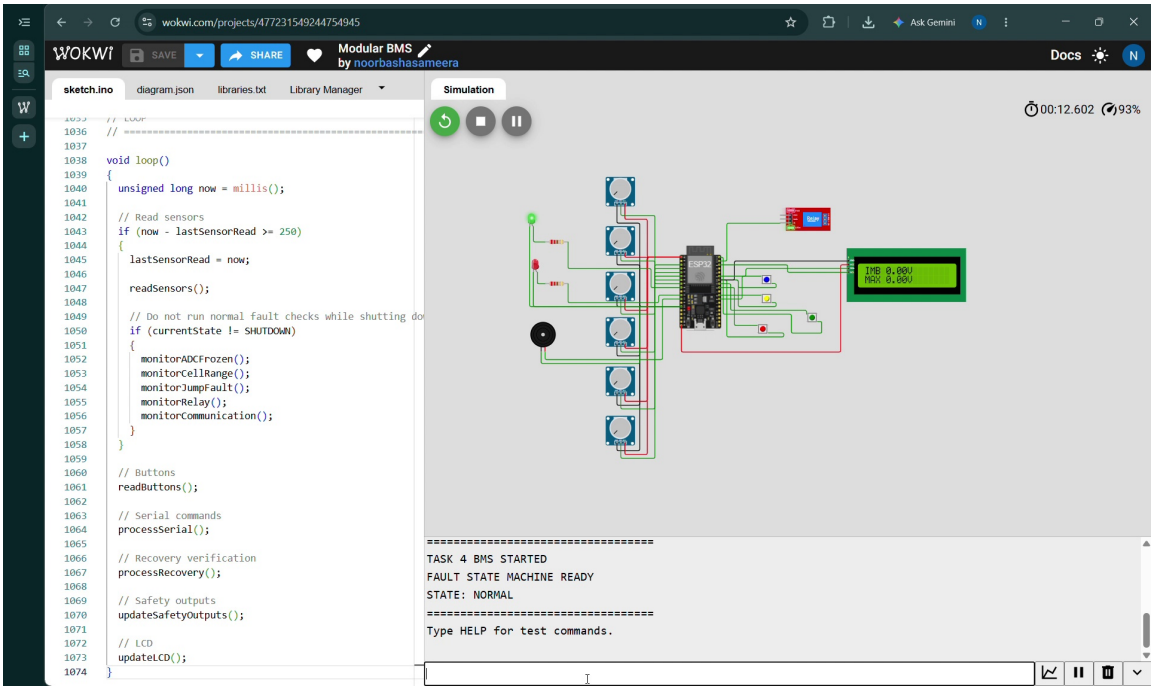


Figure 1 – NORMAL state and Task 4 state-machine startup.

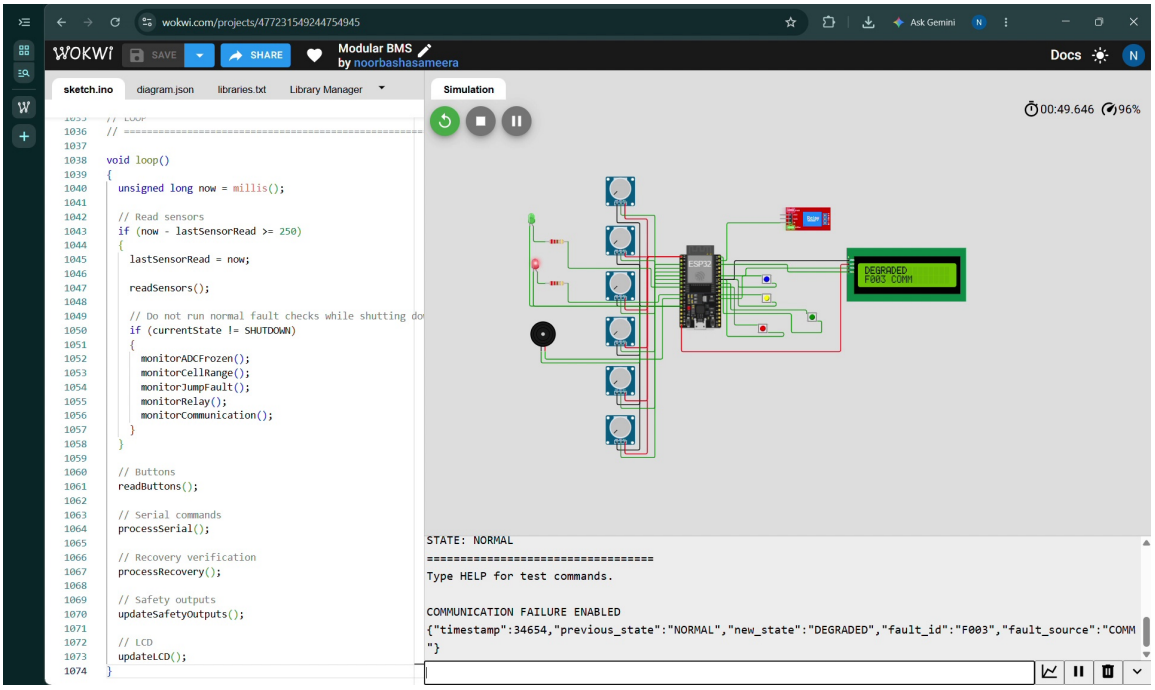


Figure 2 – Communication failure causing NORMAL → DEGRADED with F003 COMM.

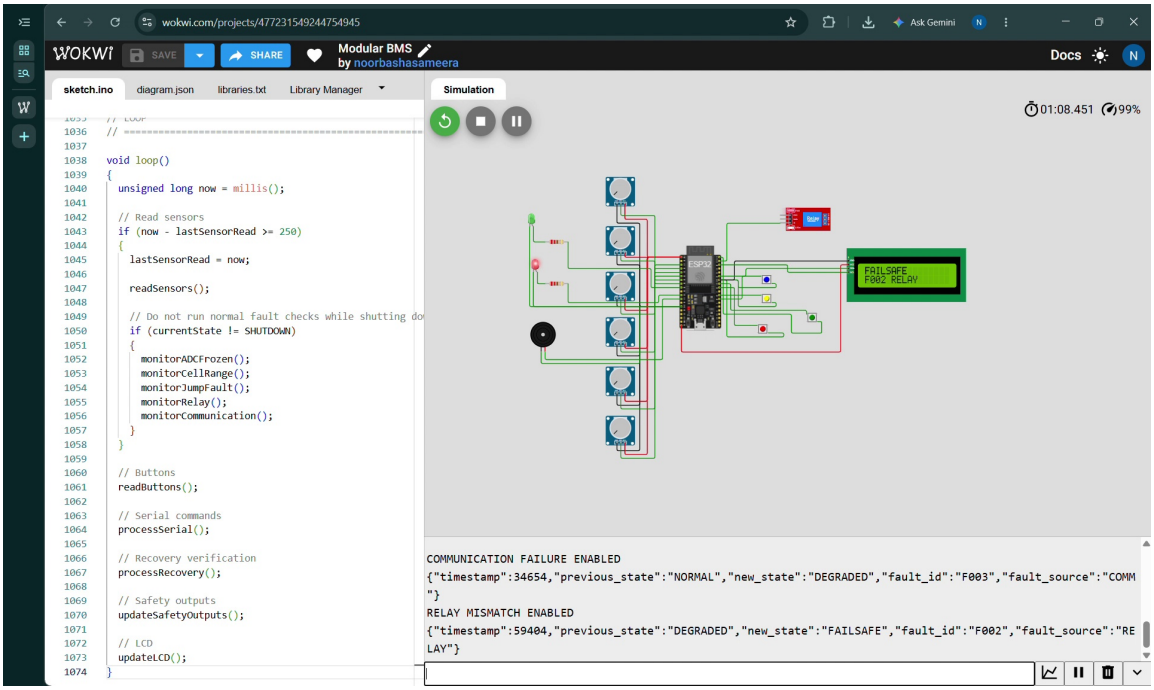


Figure 3 – Relay mismatch causing DEGRADED → FAILSAFE with F002 RELAY.

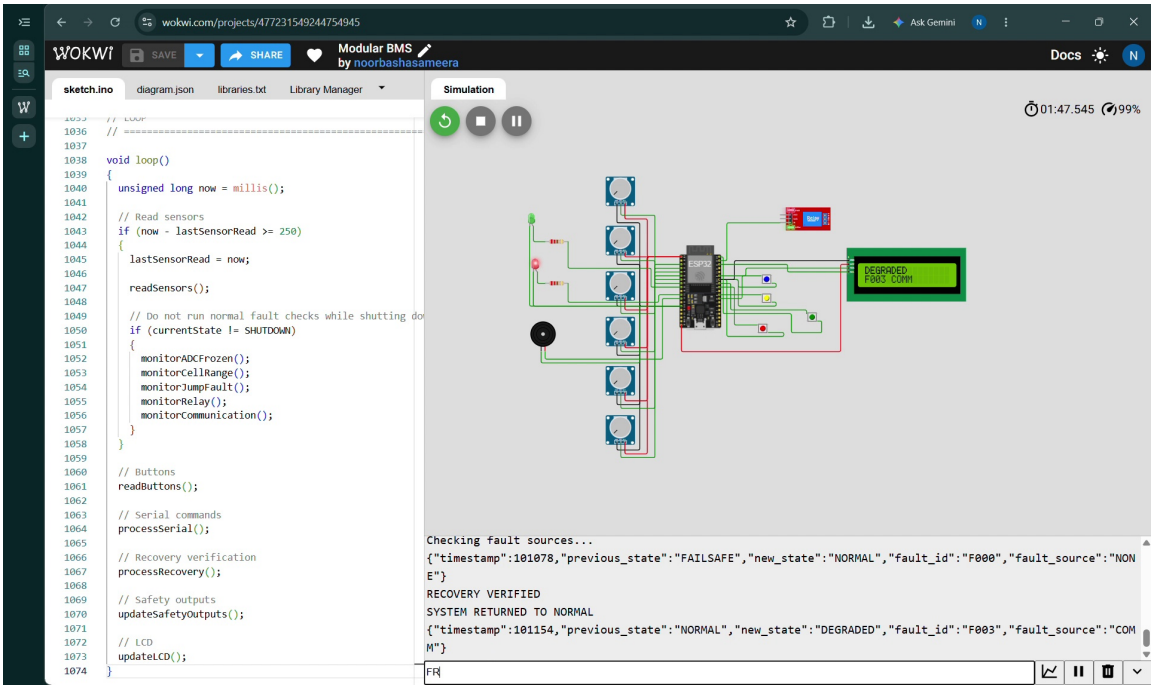


Figure 4 – Recovery verification completed and FAILSAFE → NORMAL recorded.

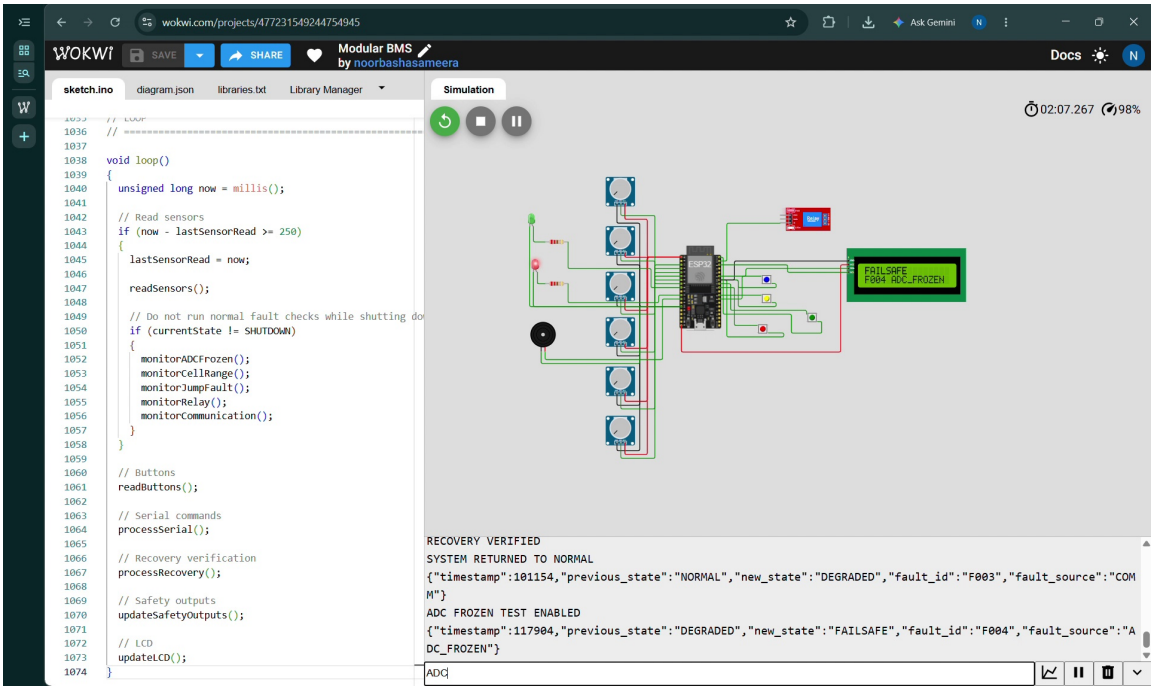


Figure 5 – Frozen ADC test causing F004 ADC_FROZEN and FAILSAFE.

9. Final Assessment

The supplied final Wokwi demonstration provides direct evidence for the core Task 4 workflow: NORMAL startup, communication degradation, relay-mismatch escalation to FAILSAFE, structured transition logging, FAILSAFE recovery verification, and frozen-ADC escalation. The circuit remains the previously working BMS circuit and the Task 4 behavior is implemented primarily through the deterministic state-machine software layer.

Submission note: The latest uploaded video is included unchanged in the submission folder. SHUTDOWN, RANGE and VOLTAGE-JUMP are documented as additional test cases, but they are not claimed as visually demonstrated in the supplied final video.

10. Submission Contents

01_Requirements – task requirement and scope
02_Circuit – pin map and Wokwi project reference
03_Code – code package note and implementation summary
04_Testing – test matrix, evidence index and screenshots
05_Report – detailed PDF report
06_Demo_Video – final uploaded Wokwi test video